COMPUTER ENGINEERING

QUEZON CITY

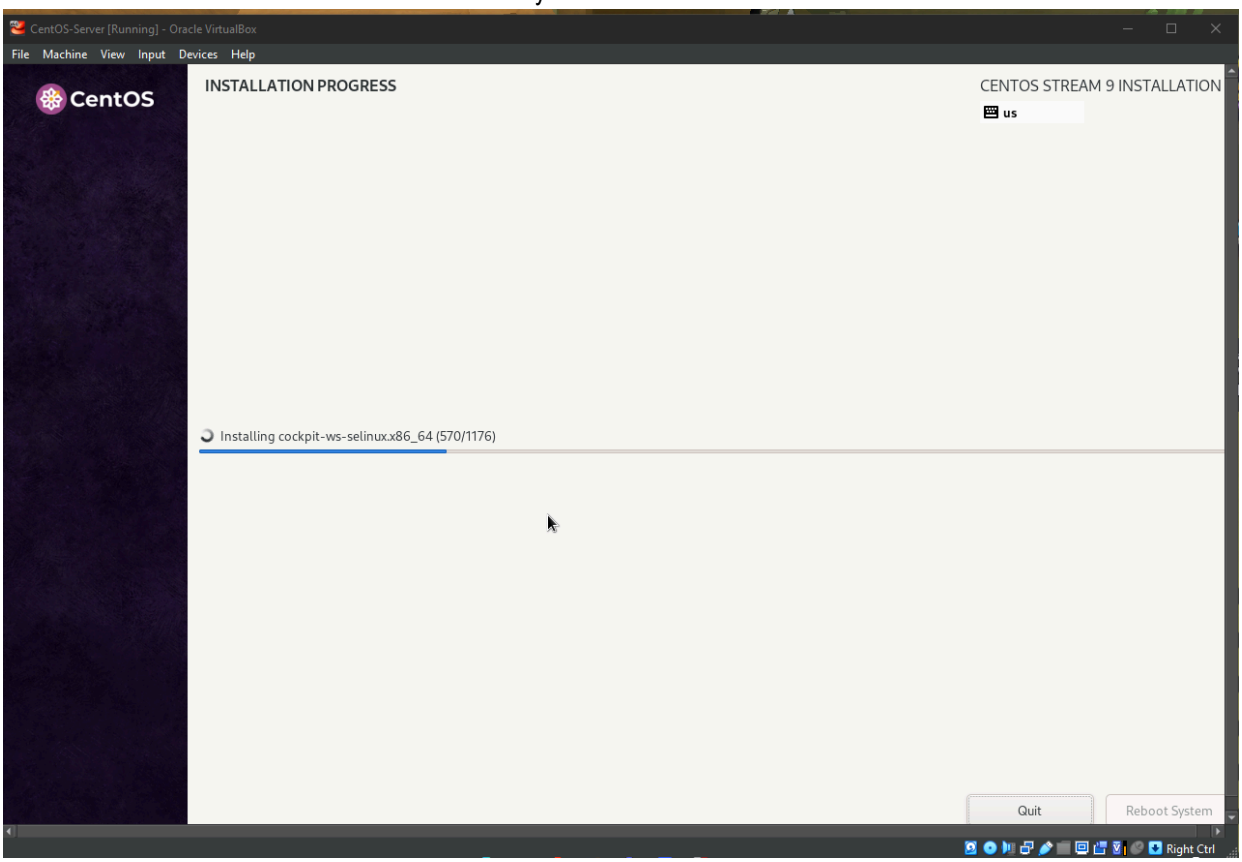
COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	07/30/26	**DATE SUBMITTED:**	08/05/26
NAME:	San Juan, Ivan Wayne	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 3 **Install SSH server on CentOS or RHEL 8**

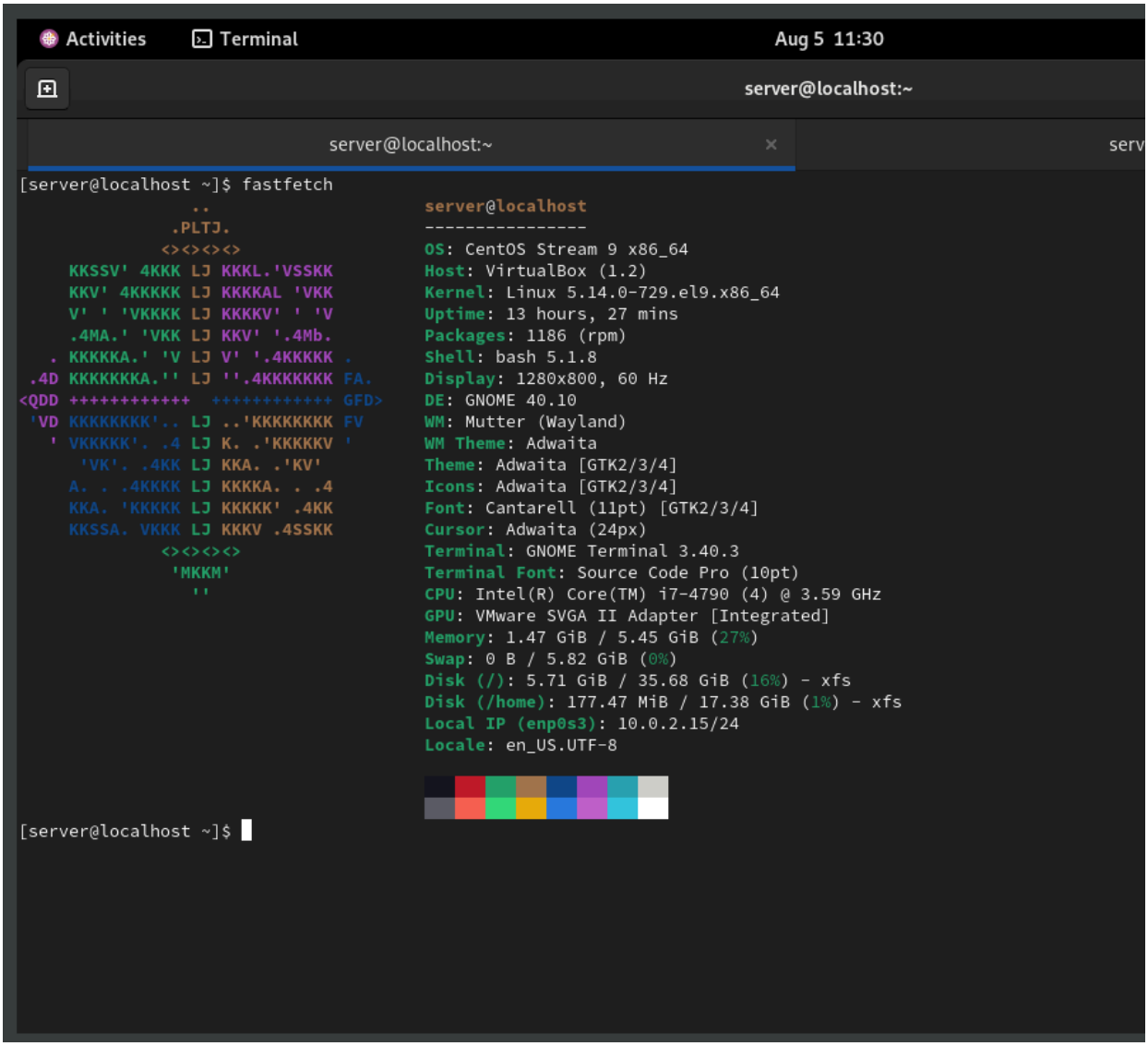
1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS 9 Stream here: https://www.centos.org/download/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



```
[server@localhost ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0   60G  0 disk
├─sda1        8:1    0    1G  0 part /boot
└─sda2        8:2    0   59G  0 part
  ├─cs-root 253:0    0 35.7G  0 lvm  /
  ├─cs-swap 253:1    0  5.8G  0 lvm  [SWAP]
  └─cs-home 253:2    0 17.4G  0 lvm  /home
sr0          11:0    1 1024M  0 rom
```



Task 2: Install the SSH server package openssh

1. Install the ssh server package openssh by using the dnf command:

$ dnf install openssh-server

```
 on most systems).
[server@localhost ~]$ sudo dnf install openssh-server
CentOS Stream 9 - BaseOS
                                                204  B/s | 3.9 kB     00:19
Errors during downloading metadata for repository 'baseos':
  - Downloading successful, but checksum doesn't match. Calculated: 86b6f10a60c
0f6032e1d480adc6406e3569cfd4392b91d298a60a435059b2477c6f0c298b4e009ab9c5134db34
e3d17aa70afdc47a8ff539043e1d527aeefc88(sha512)  Expected: 767cf48fc0ddd3971aef0
044659ced45a62e190163893301bd95aa7370b1e357f9b4dce9c21f273132551f6d02216651d293
10008b2cf99a937f572b2adc4ac1(sha512)
  - Downloading successful, but checksum doesn't match. Calculated: 17476812afd
5b2c88ea06cea3784dda7bb4a429b88bb0e0821dbd5b23d466bf0e63a4836a1aed73ce04431d1cf
e7135b1f446714c2b9224bc0cf63d6e9f8a79b(sha512)  Expected: 767cf48fc0ddd3971aef0
044659ced45a62e190163893301bd95aa7370b1e357f9b4dce9c21f273132551f6d02216651d293
10008b2cf99a937f572b2adc4ac1(sha512)
Error: Failed to download metadata for repo 'baseos': Cannot download repomd.xm
l: Cannot download repodata/repomd.xml: All mirrors were tried
[server@localhost ~]$ which ssh
/usr/bin/ssh
[server@localhost ~]$ S
```

2. Start the sshd daemon and set to start after reboot:

$ systemctl start sshd

$ systemctl enable sshd

3. Confirm that the sshd daemon is up and running:

$ systemctl status sshd

```
[server@localhost ~]$ systemctl start sshd
[server@localhost ~]$ systemctl enable sshd
[server@localhost ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: en>
     Active: active (running) since Tue 2026-08-04 22:02:45 PST; 1h 39min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1037 (sshd)
      Tasks: 1 (limit: 35273)
     Memory: 1.7M (peak: 1.9M)
        CPU: 21ms
     CGroup: /system.slice/sshd.service
             └─1037 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 04 22:02:45 localhost.localdomain systemd[1]: Starting OpenSSH server daem>
Aug 04 22:02:45 localhost.localdomain sshd[1037]: Server listening on 0.0.0.0 >
Aug 04 22:02:45 localhost.localdomain sshd[1037]: Server listening on :: port >
Aug 04 22:02:45 localhost.localdomain systemd[1]: Started OpenSSH server daemo>
...skipping...
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: en>
     Active: active (running) since Tue 2026-08-04 22:02:45 PST; 1h 39min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1037 (sshd)
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Aug 04 22:02:45 localhost.localdomain sshd[1037]: Server listening on :: port >
Aug 04 22:02:45 localhost.localdomain systemd[1]: Started OpenSSH server daemo>
~
~
```

4. Open the SSH port 22 to allow incoming traffic:

$ firewall-cmd --zone=public --permanent --add-service=ssh

$ firewall-cmd --reload

5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:

$ systemctl reload sshd

```
[server@localhost ~]$ firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[server@localhost ~]$ firewall-cmd --reload
success
[server@localhost ~]$ systemctl reload sshd
```

Task 3: Copy the Public Key to CentOS
1. Make sure that ssh is installed on the local machine.
2. Using the command ssh-copy-id, connect your local machine to CentOS.
3. On CentOS, verify that you have the authorized_keys.

```
[server@localhost ~]$ ssh server@192.168.56.104 cat ~/.ssh/authorized_keys
The authenticity of host '192.168.56.104 (192.168.56.104)' can't be established.
ED25519 key fingerprint is SHA256:RgXxs+40r5ABRazlWp5mTuC4NRNRf54hmWOMsoWGkXk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.104' (ED25519) to the list of known hosts.
server@192.168.56.104's password:
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQDb5gMAWFkPfV0QnjSnaMBe+5mOY1hTfFkhbf3PRkX5oHb0HBfhFT769WzGsalAqba
tSzOu33YO9/RFHILoYgQ4ZM3594HZrsnDAdxAEiiheLXbFUHaVA1q2L6thZI2UxLK2Kr3yvBs7FZJtyNRucKbUCdyzJLbQ15hfoecGc
C2W7q3QQsWhDKlBqYz7RdDju7YQ/Z+T8edzXDdrAm8dtbSL+GBxsiUEraQhYV2QNX5MoCSsz3FSEBolt8oOlNFtLyMWAh7eBC0Ecpq/
MvhDlquSWiUOIJzMm3D3jrWkV44kNFfuwO0QtIHuCnkGTI3IoHDbUuT2G0P4p9uqqWCSXdRt0ulHtcJ/RqwuUUOXvF3QPMAyzVayKSc
7UEtqb80O00Gqwomxhvd4cziCvtcW28HuCyN7TyaLBpSlHq4zj2n3XCLHKENQzGayUGy88Dqxl2K4DAWVjnS6ClXAQ== kaizohost@
[server@localhost ~]$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  ubuntu-files  Videos
[server@localhost ~]$
```

Task 4: Verify ssh remote connection
1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

```
kaizohost@kaizohost-VirtualBox:~$ ssh server@192.168.56.104
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Tue Aug  4 22:04:48 2026
[server@localhost ~]$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
[server@localhost ~]$ mkdir ubuntu-files
[server@localhost ~]$ ls
Desktop    Downloads  Pictures  Templates     Videos
Documents  Music      Public    ubuntu-files
[server@localhost ~]$ exit
logout
Connection to 192.168.56.104 closed.
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux Distributions?

-When choosing between Debian and Red Hat, you should focus on your primary use case, support model, and security needs, since Debian excels in community-driven development and general-purpose environments with its vast software repository and flexible release options, while Red Hat is purpose-built for enterprise production where long-term support, certified hardware compatibility, strict compliance standards, and mandatory SELinux security

are essential.

2. What are the main differences between Debian and Red Hat Linux distributions?
- The main differences between Debian and Red Hat come down to their package management systems with Debian using apt and .deb packages while Red Hat uses dnf and .rpm files, their security frameworks where Debian opts for the more permissive AppArmor while Red Hat enforces SELinux by default, and their release philosophies with Debian offering a choice between stable and rolling branches supported by community efforts while Red Hat follows a slow, point-release cycle backed by up to a decade of enterprise-grade stability updates.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I use chatgpt just to correct the grammar and I still have the ideas**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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